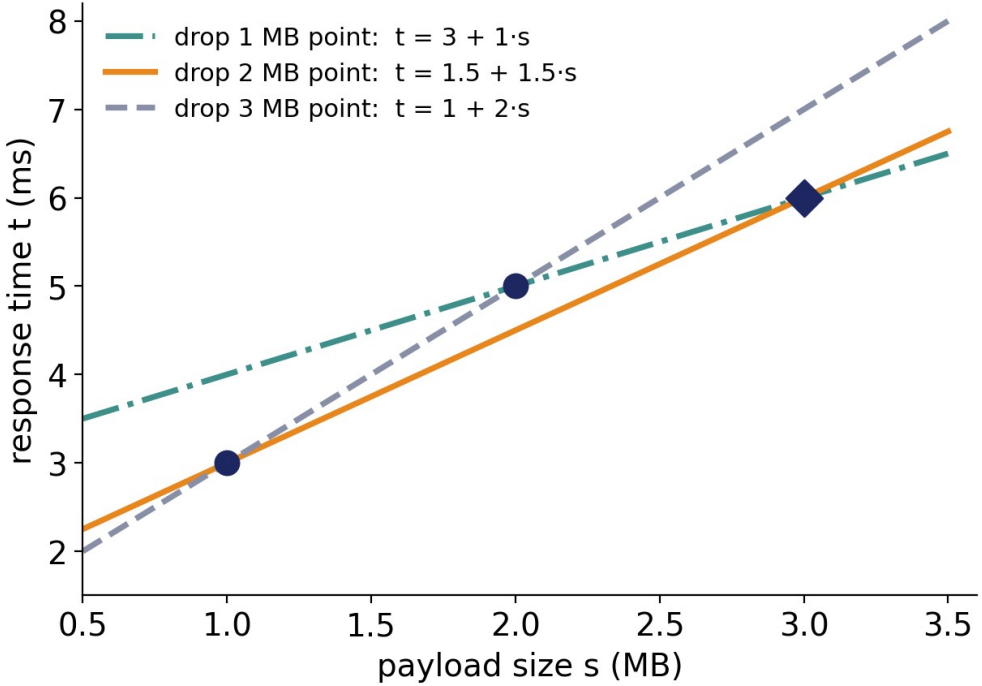
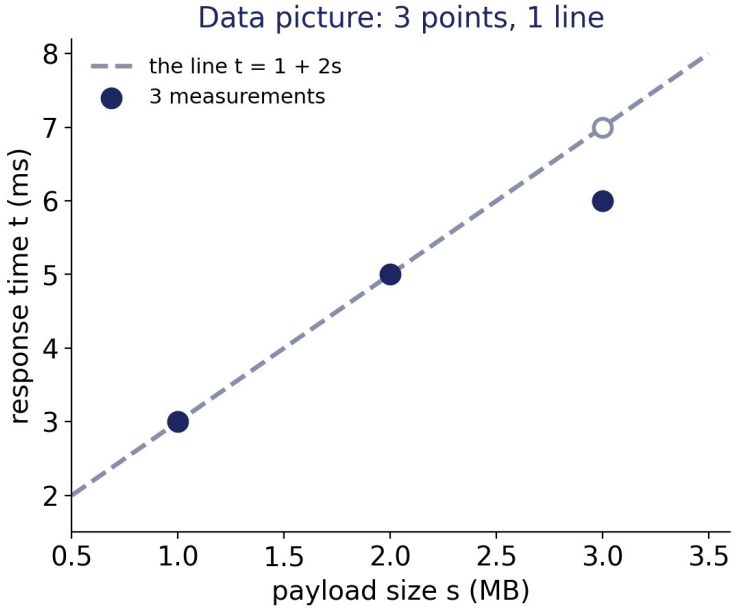
Throw one away? Which one?

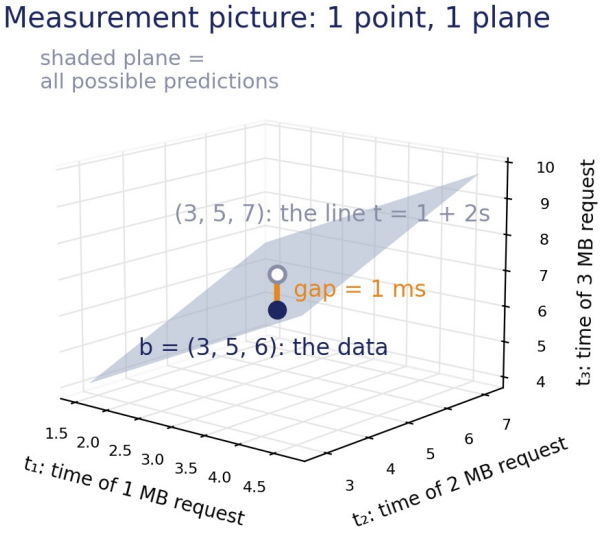


Dropped	Line	Slope
1 MB point	t = 3 + 1·s	1
2 MB point	t = 1.5 + 1.5·s	1.5
3 MB point	t = 1 + 2·s	2

Three choices, three answers — and nothing principled to pick between them. What should “the answer” mean now?

Two pictures of the same problem

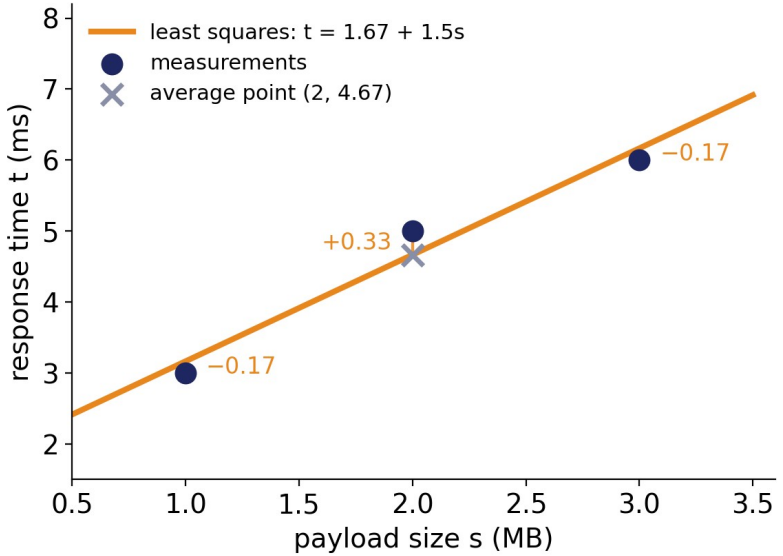




Each measurement becomes an axis. Our 3 measured times are one point, b. Every possible line gives 3 predicted times: one point on a plane.

By hand, on our three measurements

$$A^\top A = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \quad A^\top \mathbf{b} = \begin{bmatrix} 14 \\ 31 \end{bmatrix} \quad \Longrightarrow \quad \hat{\mathbf{x}} = \begin{bmatrix} 5/3 \\ 3/2 \end{bmatrix}$$



1.67 ms fixed overhead

1.5 ms per MB

0 sum of the residuals

Why do the residuals sum to exactly zero?